

PRAMITH KIRAN GOVINDA REDDY BHASKAR

Amherst, Massachusetts

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EDUCATION

University of Massachusetts Amherst

Master of Science in Computer Science

Expected Graduation: Dec 2026

Amherst, Massachusetts

Coursework: Practicum - AI, Network Security, Security Defense and Test, Advanced Information Assurance, Reinforcement Learning, Machine Learning, Software Engineering, Optimization in Computer Science.

Sri Sairam Engineering College, Chennai

B.E Computer Science and Engineering

Aug 2020 – Apr 2024

Chennai, TN, India

SKILLS

Languages: Python, C, Bash, Java, SQL, JavaScript

Security & Networking: OWASP, ELK Stack, Sigma Rules, Wireshark, Nmap, Cowrie, HASSH Fingerprinting, iptables, SSH, AES/RSA Encryption, Threat Intelligence, MITRE

Machine Learning & AI: PyTorch, TensorFlow, HuggingFace, Scikit-learn, NumPy, Pandas, HDBSCAN, Isolation Forest, LLM Evaluation, CNN

Developer Tools: Git, Docker, Kubernetes, AWS, Oracle Cloud, Linux, systemd, Flask, OpenRouter, PostgreSQL, MongoDB

Certifications: Red Hat Certified System Administrator (RHCSA), Cloud Computing (NPTEL)

EXPERIENCE

IBM Research (Thomas J. Watson Research Center)

Research Extern

Jan 2026 – Present

Amherst, Massachusetts

- Developing a LLM-as-a-Judge framework for cybersecurity via UMass Amherst COMPSCI 698DS practicum, under the supervision of Prof. Andrew McCallum (UMass Amherst) and mentored by Dr. Youngja Park (IBM), evaluating LLM responses on open-ended security questions spanning threat intelligence, vulnerability analysis, and penetration testing using a multi-dimensional CCI rubric across correctness, reasoning validity, and factual consistency.
- Constructed a unified cybersecurity benchmark of 162,000 QA pairs by collecting MITRE ATT&CK mapped datasets and public security datasets. Benchmarked 4 candidate LLMs using GPT-4.1 and Gemini 2.5 Pro as judges, achieving 0.85 Spearman's rho on human-judge agreement. Analyzed judge bias (position bias, verbosity bias) and limitations of reference-based metrics (BLEU, ROUGE) for security-domain evaluation.

Vectra Technosoft Private Limited

Software Engineering Intern

Mar 2023 – May 2023

Chennai, India

- Containerized applications using Podman and Docker on Red Hat Linux, streamlining deployment workflows and ensuring environment consistency. Configured virtual networks and administered Linux system services to support multi-container environments.
- Earned Red Hat Certified System Administrator (RHCSA) certification through hands-on work with Red Hat Linux during internship, validating proficiency in system administration and Linux environment management.

PROJECTS

HoneyIntel | Python, Cowrie, ELK Stack, Sigma Rules, IDS

Jan 2026 – Present

- Deployed live SSH honeypot on Oracle Cloud (Cowrie), collecting 32,800 attacker sessions across 59 HASSH fingerprint groups and enriching IPs with threat intelligence APIs (AbuseIPDB, Shodan, VirusTotal, and ip-api) to identify malicious infrastructure and credential spraying patterns.
- Built a host-based IDS and automated detection and response system (systemd and Telegram) that monitors honeypot health, auto-blocks brute-force attackers via iptables, alerts on unauthorized open ports, and triggers emergency shutdown on confirmed IOCs, validated through 8 real incidents.
- Performed log analysis using HDBSCAN clustering and Isolation Forest anomaly detection to triage sessions into 34 behavioral clusters, flagging 981 high-priority anomalies. Integrating ELK Stack and Sigma detection rules per cluster for SIEM-compatible alerting.

Aero Encrypt | Python, Flutter, Docker

2022 – 2024

- Designed and developed offline-first password manager using Flutter to address security risks of cloud-based managers, storing credentials locally with hybrid Advanced Encryption Standard (AES) + RSA encryption, ensuring no password data leaves device.
- Integrated Firebase solely for user authentication, keeping credential storage fully on-device. Led team end-to-end from architecture design through deployment as primary developer.

Lung Cancer Detection System | Python, CNN, Watershed Algorithm

2021 – 2024

- Analyzed Computed Tomography (CT) scans using Watershed Algorithm for feature extraction and Convolutional Neural Network (CNN) for classification, achieving significant improvement in early-stage diagnostic accuracy through web-based interface.
- Co-authored and presented findings at ICCEBS'23 (published on IEEE Xplore), demonstrating real-world applicability for assisting healthcare professionals in medical imaging workflows.